

A Strict Proof of the Even Minimal-Solution Anchor $K(1) = e/4$

BVE research project

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Abstract

The repository previously recorded the even minimal-solution constant $K(1) = e/4$ as a high-precision numerical conjecture. This note proves the claim directly from the defining backward recurrence. An exact factorial scaling reduces the third-order recurrence to a first-order recurrence for a second difference. The finite backward solution, its fixed-index limit, the unique minimal branch, and the asymptotic tail are all obtained by positive factorial series. No numerical fit, formal asymptotic ansatz, or unchecked computer algebra is used.

1 Statement

For $j \geq 3$, set

$$\begin{aligned} P_j &= 8j^2 - 4j + \frac{j}{j-1}, \\ Q_j &= 4j(j-1)(2j-1)(2j-3) + 4j(2j-3), \\ R_j &= 4j(j-2)(2j-3)(2j-5). \end{aligned}$$

Consider

$$\mu_j = P_j \mu_{j-1} - Q_j \mu_{j-2} + R_j \mu_{j-3}. \quad (\text{R})$$

For $N \geq 3$, define the backward solution by

$$\mu_{N+1}^{(N)} = 1, \quad \mu_N^{(N)} = \mu_{N-1}^{(N)} = 0,$$

and use (R) backwards for $j = N+1, N, \dots, 3$. Whenever the denominator is nonzero, put $\hat{\mu}_k^{(N)} = \mu_k^{(N)} / \mu_0^{(N)}$.

Theorem 1 (Strict $K(1)$ anchor) *For every fixed $k \geq 0$, the limit $\mu_k^* = \lim_{N \rightarrow \infty} \hat{\mu}_k^{(N)}$ exists. It is the unique normalized minimal solution and has the exact representation*

$$\mu_k^* = 2e(2k)! \sum_{r=k+2}^{\infty} \frac{r-k-1}{(2r-1)!}. \quad (1)$$

Moreover,

$$\lim_{j \rightarrow \infty} j^3 \mu_j^* = \frac{e}{4}. \quad (2)$$

2 Exact factorization

Define

$$v_j = \frac{\mu_j}{(2j)!}, \quad c_j = \frac{1}{2(j-1)(2j-1)}.$$

Direct cancellation gives

$$\frac{P_j(2j-2)!}{(2j)!} = 2 + c_j, \quad \frac{Q_j(2j-4)!}{(2j)!} = 1 + 2c_j, \quad \frac{R_j(2j-6)!}{(2j)!} = c_j.$$

Thus (R) is equivalent to

$$v_j = (2 + c_j)v_{j-1} - (1 + 2c_j)v_{j-2} + c_jv_{j-3}. \quad (3)$$

Set

$$d_j = v_j - 2v_{j-1} + v_{j-2}.$$

Equation (3) factors as

$$d_j = c_j d_{j-1}. \quad (4)$$

Since

$$\frac{j/(2j)!}{(j-1)/(2j-2)!} = \frac{1}{2(j-1)(2j-1)} = c_j,$$

all solutions of (4) have the form

$$d_j = C \frac{j}{(2j)!} \quad (5)$$

for a constant C .

3 Finite backward solutions and the fixed-index limit

For the terminal solution, $v_N^{(N)} = v_{N-1}^{(N)} = 0$ and $v_{N+1}^{(N)} = 1/(2N+2)!$. Hence

$$d_N^{(N)} = 0, \quad d_{N+1}^{(N)} = \frac{1}{(2N+2)!}.$$

Backward use of (4) gives

$$d_j^{(N)} = \frac{j}{(N+1)(2j)!}, \quad 2 \leq j \leq N+1. \quad (6)$$

Twice summing the second differences in (6), with the two terminal zeros, gives the exact finite formula

$$\mu_j^{(N)} = \frac{(2j)!}{2N+2} \sum_{r=j+2}^N \frac{r-j-1}{(2r-1)!}, \quad 0 \leq j \leq N-1. \quad (7)$$

In particular,

$$\mu_0^{(N)} = \frac{1}{2N+2} \sum_{r=2}^N \frac{r-1}{(2r-1)!} > 0$$

for every $N \geq 3$. Therefore the prescribed normalization is always valid, not merely eventually.

For fixed k , (7) yields

$$\hat{\mu}_k^{(N)} = (2k)! \frac{\sum_{r=k+2}^N (r-k-1)/(2r-1)!}{\sum_{r=2}^N (r-1)/(2r-1)!}.$$

Both sums are positive and convergent. Moreover,

$$\begin{aligned} \sum_{r=2}^{\infty} \frac{r-1}{(2r-1)!} &= \frac{1}{2} \sum_{r=2}^{\infty} \left(\frac{1}{(2r-2)!} - \frac{1}{(2r-1)!} \right) \\ &= \frac{1}{2} ((\cosh 1 - 1) - (\sinh 1 - 1)) = \frac{1}{2e}. \end{aligned}$$

Taking the limit proves (1) and $\mu_0^* = 1$.

4 Minimality and uniqueness

Define

$$\phi_j = \sum_{r=j+2}^{\infty} \frac{r(r-j-1)}{(2r)!}.$$

The series is absolutely convergent and tends to zero. Direct telescoping of the tails gives

$$\phi_j - 2\phi_{j-1} + \phi_{j-2} = \frac{j}{(2j)!}.$$

Consequently every solution of (3) has the form

$$\frac{\mu_j}{(2j)!} = A + Bj + C\phi_j. \quad (8)$$

The minimal branch is the subfactorial branch, equivalently the branch for which $\mu_j/(2j)! \rightarrow 0$. By (8), this forces $A = B = 0$, so the minimal solution space is one-dimensional. Since

$$\phi_0 = \sum_{r=2}^{\infty} \frac{r(r-1)}{(2r)!} = \frac{1}{4e},$$

the condition $\mu_0 = 1$ fixes $C = 4e$. The resulting formula is exactly (1), so the backward limit is the unique normalized minimal solution.

5 Asymptotic constant

Write $r = j + \ell + 1$ in (1):

$$\mu_j^* = 2e \sum_{\ell=1}^{\infty} \ell \frac{(2j)!}{(2j+2\ell+1)!}. \quad (9)$$

The first term is

$$T_{j,1} = \frac{2e}{(2j+1)(2j+2)(2j+3)}.$$

For $\ell \geq 2$, every extra factorial factor is at least $2j + 1$, so

$$0 \leq j^3 \sum_{\ell=2}^{\infty} \ell \frac{(2j)!}{(2j+2\ell+1)!} \leq j^3 x_j^5 \left(\frac{2}{1-x_j^2} + \frac{x_j^2}{(1-x_j^2)^2} \right), \quad x_j = \frac{1}{2j+1}.$$

The right side is $O(j^{-2})$. Finally,

$$j^3 T_{j,1} \longrightarrow \frac{e}{4},$$

which proves (2).

6 Index and sign audit

At the last backward step,

$$P_3 = \frac{123}{2}, \quad Q_3 = 396, \quad R_3 = 36, \quad c_3 = \frac{1}{20},$$

so every required denominator is nonzero. The upper terminal check from (7) is

$$\mu_{N-2}^{(N)} = \frac{1}{R_{N+1}},$$

which makes the $j = N + 1$ equation recover $\mu_{N+1}^{(N)} = 1$ with the prescribed zeros at N and $N - 1$. For $N = 3$, (7) gives $\mu_1^{(3)} = 1/480$ and $\mu_0^{(3)} = 11/480$, agreeing with direct backward substitution.

All rearrangements above involve nonnegative terms. No numerical evidence, formal asymptotic ansatz, or unchecked symbolic computation is used in the proof.

Remark 2 *This theorem settles the $c = 1$ anchor only. A closed form for the general constant $K(c)$, the source-term control in the broader recurrence program, and the general coefficient-family classification remain open.*